Lab Assignment

Task 1: Configure the Web App Manifest

Create a valid `manifest.json` file in your application root directory. Define the application's identity by including the following properties: `name`, `short_name`, `start_url`, `display` (set to `standalone`), `theme_color`, and an `icons` array containing paths to icons of at least 192x192px and 512x512px sizes. Link this manifest file in the `<head>` section of your `index.html` file.

Task 2: Register the Service Worker

Write a script in your main client-side JavaScript file to register your service worker. Wrap the registration code inside a check for feature support ('serviceWorker' in navigator) and ensure it registers the sw.js file safely after the window has fully loaded to prevent blocking the initial page paint.

Task 3: Implement Static Caching on Install

Inside your sw.js file, add an event listener for the install lifecycle event. Create a cache named v1-static-assets using the Cache Storage API, and use event.waitUntil() along with cache.addAll() to pre-cache your core application shell files, including your main index.html, basic CSS stylesheets, core JavaScript files, and an offline fallback page

Task 4: Clean Up Old Caches on Activate

Add an event listener for the activate lifecycle event inside your sw.js file. Use this event to clear out older versions of your cache. Write logic that loops through all existing cache keys, checks if they match your current cache name, and uses caches.delete(key) to drop old caches so that updated assets take effect immediately.

Task 5: Intercept Network Requests for Offline Fallback

Add a fetch event listener to your sw.js file to intercept outgoing network requests. Implement a "Cache, falling back to Network" strategy for your static assets. If a request fails completely because the browser is offline, catch the network error and serve your pre-cached offline fallback page instead.